Counselling Code : VBIT

VIGNANA BHARATHI Institute of Technology

(A UGC Autonomous Institution, Approved by AICTE, Accredited by NBA & NAAC-A Grade, Affiliated to JNTUH)

SKILL DEVELOPMENT COURSE – UI DESIGN FLUTTER

LAB PROJECT

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Year/ Semester : III/ I

Branch & Section : CSE – (B)

NAVIGATION IN FLUTTER

OVERVIEW

This project, Navigation in Flutter, focuses on implementing smooth and visually appealing transitions between multiple pages within a Flutter mobile application. The goal is to explore and demonstrate various navigation techniques such as named routes, push/pop navigation, and animated page transitions, while enhancing the user experience through Hero animations and dynamic visual effectsThe project includes multiple colorful pages, each designed with animated floating objects, gradient backgrounds, and glass-morphism cards to create an engaging and modern UI. Flutter's Navigator and PageRouteBuilder are utilized to provide customized animations like fade, scale, and slide transitions, ensuring smooth and natural movement between screens.The Hero animation is implemented to create a seamless shared-element transition, where common visual elements (like images or logos) smoothly transform between pages, providing a professional and interactive look. The project also showcases the use of animation controllers for continuous floating motion effects and stack-based layout design for layering visuals beautifully.By combining powerful navigation mechanisms with advanced UI animations, this project demonstrates how Flutter can be used to build rich, interactive, and visually appealing applications that enhance user engagement and overall app experience.

CODE :

```
import 'package:flutter/material.dart';
import 'dart:math' as math;
void main() => runApp(NavigationDemoApp());
class NavigationDemoApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Animated Navigation Demo',
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.deepPurple),
        useMaterial3: true,
      ),
      home: HomePage(),
    );
  }
}
class HomePage extends StatefulWidget {
  @override
  _HomePageState createState() => _HomePageState();
}
class _HomePageState extends State<HomePage> with TickerProviderStateMixin {
  late AnimationController _circleController;
  late AnimationController _rotateController;
 @override
  void initState() {
    super.initState();
    _circleController = AnimationController(vsync: this, duration: const Duration(seconds: 8))
      ..repeat();
    _rotateController = AnimationController(vsync: this, duration: const Duration(seconds:
          12))
```

```
repeat();
  }
 @override
  void dispose() {
    _circleController.dispose();
    _rotateController.dispose();
    super.dispose();
  }
Widget _floatingCircle(Color color, double radius, double offsetX, double offsetY) {
return AnimatedBuilder(
      animation: _circleController,
      builder: (context, child) {
        double dx = offsetX + 50 * math.sin(_circleController.value * 2 * math.pi);
        double dy = offsetY + 40 * math.cos(_circleController.value * 2 * math.pi);
        return Transform.translate(offset: Offset(dx, dy), child: child);
      },
      child: Container(
        width: radius,
        height: radius,
        decoration: BoxDecoration(
          color: color.withOpacity(0.5),
          shape: BoxShape.circle,
          boxShadow: [
            BoxShadow(color: color.withOpacity(0.5), blurRadius: 20, spreadRadius: 10),
          ],
        ),
      ),
    );
  }
Widget _rotatingRing() {
    return AnimatedBuilder(
      animation: _rotateController,
```

```
      builder: (context, child) {
        return Transform.rotate(
          angle: _rotateController.value * 2 * math.pi,
          child: child,
        );
      },
      child: Container(
        width: 180,
        height: 180,
        decoration: BoxDecoration(
          border: Border.all(color: Colors.white.withOpacity(0.6), width: 3),
          shape: BoxShape.circle,
        ),
      ),
    );
  }
@override
  Widget build(BuildContext context) {
    return Scaffold(
      body: Stack(
        children: [
          // Gradient background
          Container(
            decoration: const BoxDecoration(
              gradient: LinearGradient(
                colors: [Color(0xFF8E2DE2), Color(0xFF4A00E0), Color(0xFFFF6FB5)],
                begin: Alignment.topLeft,
                end: Alignment.bottomRight,
              ),
            ),
          ),
          Positioned(top: 100, left: 40, child: _floatingCircle(Colors.pinkAccent, 100, 30, 30)),
```

```
Positioned(bottom: 100, right: 60, child: _floatingCircle(Colors.blueAccent, 80, 10,
  10)),
Positioned(bottom: 200, left: 70, child: _floatingCircle(Colors.orange, 60, 20, 20)),
Center(child: _rotatingRing()),
Center(
  child: Column(
    mainAxisAlignment: MainAxisAlignment.center,
    children: [
      Hero(
        tag: 'hero-logo',
        child: CircleAvatar(
          radius: 65,
          backgroundImage: NetworkImage(
            'https://flutter.dev/assets/homepage/carousel/slide_1-bg-opaque-2a2c.png',
          ),
        ),
      ),
      const SizedBox(height: 50),
      ElevatedButton(
        style: ElevatedButton.styleFrom(
          backgroundColor: Colors.white,
          foregroundColor: Colors.deepPurple,
          padding: const EdgeInsets.symmetric(horizontal: 50, vertical: 15),
          shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(25)),
        ),
        onPressed: () {
          Navigator.push(
            context,
            PageRouteBuilder(
              transitionDuration: const Duration(milliseconds: 800),
              pageBuilder: (_, __, ___) => SecondPage(),
              transitionsBuilder: (_, animation, __, child) => FadeTransition(
```

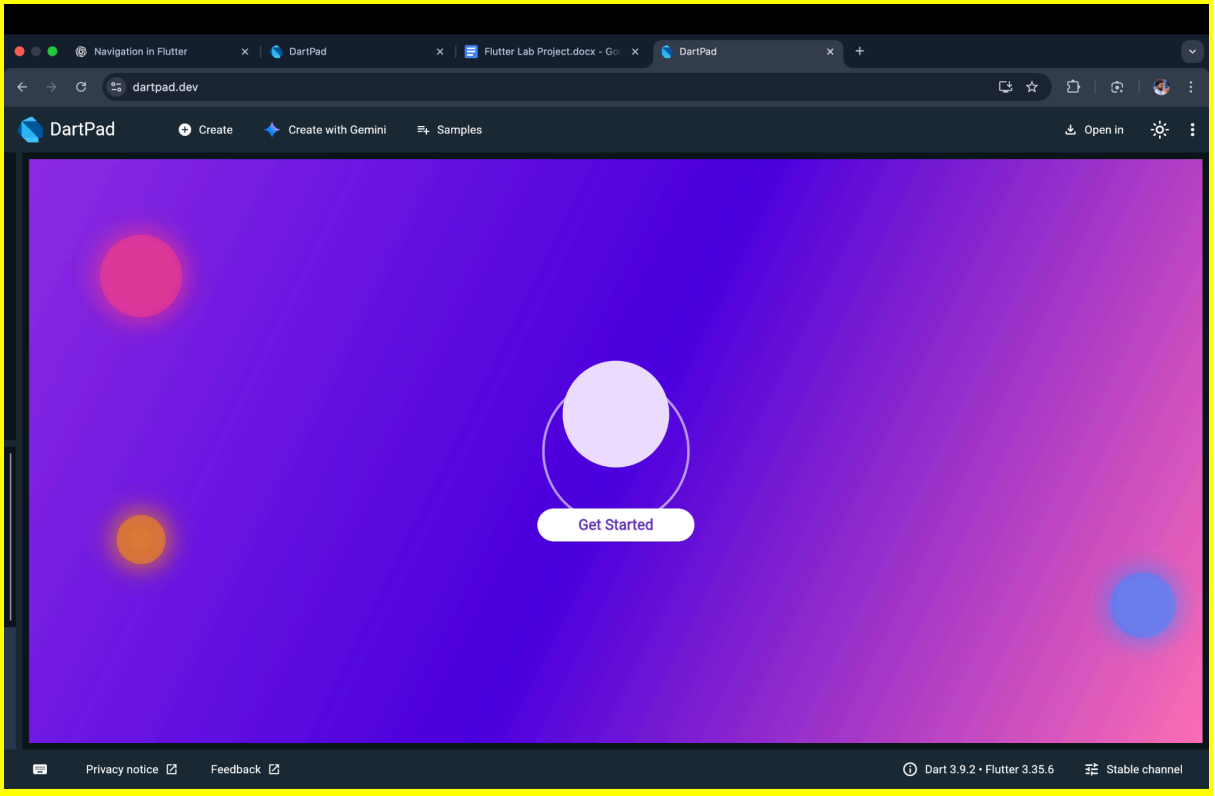
```
                opacity: animation,
                child: ScaleTransition(
                  scale: Tween(begin: 0.9, end: 1.0).animate(animation),
                  child: child,
                ),
              ),
            ),
          );
        },
        child: const Text("Get Started",
            style: TextStyle(fontSize: 18, fontWeight: FontWeight.bold)),
      ),
    ],
  ),
),
],
),
);
}
}
class SecondPage extends StatefulWidget {
  @override
  _SecondPageState createState() => _SecondPageState();
}
class _SecondPageState extends State<SecondPage> with TickerProviderStateMixin {
  late AnimationController _floatController;
@override
  void initState() {
    super.initState();
    _floatController = AnimationController(vsync: this, duration: const Duration(seconds: 10))
      ..repeat();
  }
```

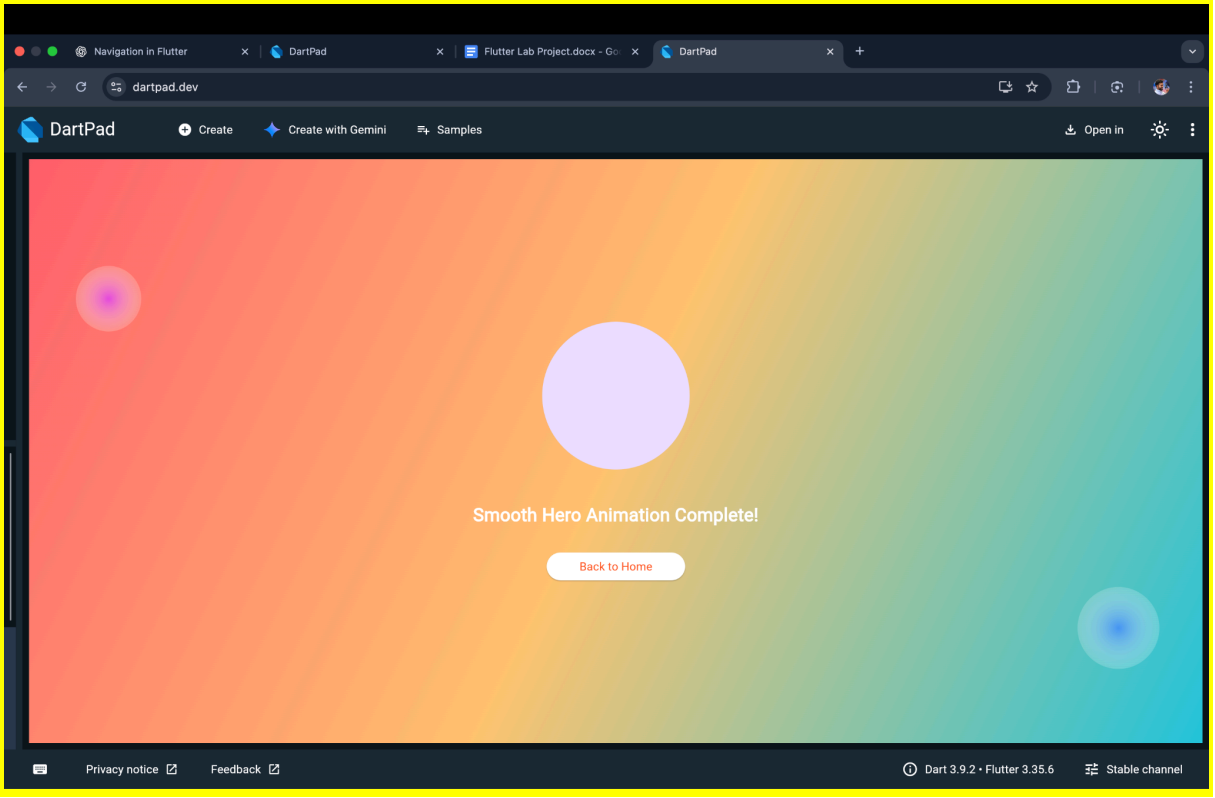
```
  @override
  void dispose() {
    _floatController.dispose();
    super.dispose();
  }
Widget _floatingShape(Color color, double size, double offsetX, double offsetY) {
    return AnimatedBuilder(
      animation: _floatController,
      builder: (context, child) {
        double dx = offsetX + 40 * math.sin(_floatController.value * 2 * math.pi);
        double dy = offsetY + 30 * math.cos(_floatController.value * 2 * math.pi);
        return Transform.translate(offset: Offset(dx, dy), child: child);
      },
      child: Container(
        width: size,
        height: size,
        decoration: BoxDecoration(
          shape: BoxShape.circle,
          gradient: RadialGradient(
            colors: [color.withOpacity(0.8), Colors.white.withOpacity(0.2)],
          ),
        ),
      ),
    );
  }
@override
  Widget build(BuildContext context) {
    return Scaffold(
      body: Stack(
        children: [
          // Background gradient
```

```
  Container(
    decoration: const BoxDecoration(
      gradient: LinearGradient(
        colors: [Color(0xFFFF5F6D), Color(0xFFFFC371), Color(0xFF24C6DC)],
        begin: Alignment.topLeft,
        end: Alignment.bottomRight,
      ),
    ),
  ),
  // Floating shapes
  Positioned(top: 100, left: 60, child: _floatingShape(Colors.purpleAccent, 80, 0, 0)),
  Positioned(bottom: 120, right: 50, child: _floatingShape(Colors.blueAccent, 100, 0,
    0)),
Center(
    child: Column(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [
        Hero(
          tag: 'hero-logo',
          child: CircleAvatar(
            radius: 90,
            backgroundImage: NetworkImage(
                'https://flutter.dev/assets/homepage/carousel/slide_1-bg-opaque-2a2c.png'),
          ),
        ),
        const SizedBox(height: 40),
        const Text(
          "Smooth Hero Animation Complete!",
          textAlign: TextAlign.center,
          style: TextStyle(
              fontSize: 22, fontWeight: FontWeight.bold, color: Colors.white),
        ),
```

```
            const SizedBox(height: 30),
            ElevatedButton(
              style: ElevatedButton.styleFrom(
                backgroundColor: Colors.white,
                foregroundColor: Colors.deepOrange,
                padding: const EdgeInsets.symmetric(horizontal: 40, vertical: 15),
                shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(25)),
              ),
              onPressed: () => Navigator.pop(context),
              child: const Text('Back to Home'),
            ),
          ],
        ),
      ),
    ],
  ),
);
}
}
```

OUTPUT :


Navigation in Flutter
DartPad
Flutter Lab Project.docx - Go
DartPad
dartpad.dev
DartPad
Create
Create with Gemini
Samples
Open in
Get Started
Privacy notice
Feedback
Dart 3.9.2 • Flutter 3.35.6
Stable channel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